A Systematic Review of Medicaid Healthcare Utilization by Adults with Serious Mental Illness

Christine Langston
Ann Kutney-Lee
Matthew McHugh
J. Margo Brooks Carthon

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Langston, C., Kutney-Lee, A., McHugh, M., & Carthon, J.M.B. (2025). A systematic review of Medicaid healthcare utilization by adults with serious mental illness. *MEDSURG Nursing, 34*(2), 75-85. https://doi.org/10.62116/MSJ.2025.34.2.75

Medicaid-insured patients with serious mental illness experience higher healthcare costs and service utilization, including increased rates of inpatient hospitalizations, emergency department visits, outpatient visits, and 30-day readmissions. The current review addresses healthcare utilization outcomes among this population.

Keywords: serious mental illness, Medicaid, healthcare resource utilization

Statement of Clinical Problem

Up to 14.2 million U.S. adults are living with a serious mental illness (SMI, e.g., schizophrenia, psychosis, bipolar disorder, major depressive disorder). The quality of health care they receive varies significantly based on insurance payer type (McConnell et al., 2020; Mongelli et al., 2020). Up to 50% of Medicaid-insured persons with SMI reported not receiving needed health care in 2018; when care is offered, studies suggest patients are less likely to receive guideline-concordant care than their commercially insured counterparts (Gross et al., 2022; Medicaid and CHIP Payment and Access Commission, 2021). For example, persons living with SMI have higher rates of diabetes and cardiovascular risk factors but are less likely to receive recommended screenings and interventions. These disparities are even more pronounced among patients with SMI who are insured by Medicaid (Hall et al., 2019; Hwong et al., 2022). A systematic review by Ronaldson and co-authors (2020) found these patients also are more likely to experience an increase in healthcare resource utilization (HRU) without any increase in the quality of the health care they receive.

Patterns of HRU among Medicaid patients with SMI are likely multifactorial, linked to factors operating at the level of the individual, providers, and healthcare systems (Recchia et al., 2022) The relationship between Medicaid insurance and welfare has resulted in significant social stigma, stringent eligibility requirements that

Christine Langston, BSN, RN, is Pre-Doctoral Fellow, Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, PA.

Ann Kutney-Lee, PhD, MSN, RN, FAAN, is Adjunct Associate Professor, Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing; Associate Director of Analytics and Methods, Center for Health Equity and Research Promotion, Corporal Michael J. Crescenz VA Medical Center, Philadelphia, PA.

Matthew McHugh, PhD, JD, MPH, RN, FAAN, is Professor of Nursing and Director, Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, PA.

J. Margo Brooks Carthon, PhD, APRN, FAAN, is Professor of Nursing and Associate Director, Center for Health Outcomes and Policy Research, University of Pennsylvania School of Nursing, Philadelphia, PA.

Acknowledgment: The authors would like to acknowledge Richard James, Nursing Liaison Librarian, University of Pennsylvania Biomedical Library.

Note: This research was supported by funding from the National Institutes of Nursing Research Grant No. T32NR007104 Linda Aiken, PI, and the Hillman Scholars Program in Nursing Innovation. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

discourage enrollment, and lower standards of care for the Medicaid-insured (Singletary & Chin, 2023). Low-income status itself is associated with challenges in accessing care (McMaughan et al., 2020). In 2023, many states set eligibility at 138% of the federal poverty level, which translated to an annual income of about $20,120 for an individual and $41,400 for a family of four (Kaiser Family Foundation, 2023).

As one of the primary drivers of health disparities and a well-established social determinant of health, economic disadvantage has been associated with difficulties accessing transportation, concerns over medical co-pay, low health literacy, and distrust of healthcare providers (Brom et al., 2022). Persons with low income, including those insured by Medicaid, also have expressed experiences of discrimination when accessing healthcare services (Alcala et al., 2020; Nong et al., 2020). These experiences may be heightened given people of color are insured disproportionately by Medicaid (Hill et al., 2025; Snowden & Graaf, 2019). According to American Community Survey coverage data for 2008-2021 reported by the Kaiser Family Foundation (2023), Hispanic Americans, African Americans, and American Indian/Alaska Native Americans accounted for 31.3%, 34.5% and 38.1% of Medicaid enrollees, respectively.

Objectives of the Review

This systematic review addresses healthcare utilization outcomes among Medicaid-insured patients with SMI. The review summarizes and appraises the quality of literature related to disparities in healthcare utilization and outcomes for Medicaid-insured patients with SMI, identifies factors that help to explain disparate outcomes, and assesses how nursing care may be tailored to meet the needs of these patients at risk for inequitable health outcomes.

Relevant Literature

Disparities in HRU patterns and access to healthcare services are common among Medicaid-insured patients, but it remains unclear how healthcare providers and staff might influence these disparate outcomes (Gandhi et al., 2021; Rajeswaren et al., 2024). Nurses in particular use important knowledge skillsets, such as assessment, diagnosis, and treatment, to provide mental and physical health care to patients with SMI in acute, post-acute, and outpatient settings (van Dusseldorp et al., 2023; Yalcin et al., 2019).

The National Academies of Sciences, Engineering, and Medicine (NASEM, 2021) Committee on the Future of Nursing 2020-2030 called for all nurses at all levels and in all healthcare settings to embrace their role in achieving true health equity by working with other healthcare professionals and sectors to address underlying social determinants of health that lead to health disparities for vulnerable populations. However, it is unclear if the role of nurses is considered in current literature related to disparities among low-income Medicaid-insured patients with SMI and if there are implications across existing studies for nursing care practice.

High-quality nursing care and nurse-led interventions such as increased screening for social needs among at risk patients have been shown to mediate healthcare disparities in other vulnerable populations (American Nurses Association, 2018; Brooks Carthon et al., 2021; NASEM, 2021). Despite the evidence pointing to the pivotal role of nursing care in the health outcomes of patients with SMI, there is a noticeable lack of literature focused on developing targeted, efficacious nursing interventions specifically for Medicaid-insured patients with SMI.

Methods for Conducting the Review

Conduct and reporting of this systematic review were guided by the standards of Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Statement (Page et al., 2021). First, a systematic review was conducted using key search terms to identify relevant studies. Selected studies then were evaluated against inclusion criteria. Quality of included studies was appraised critically, and a narrative synthesis of reported results of these studies was conducted.

A search of peer-reviewed literature published January 2017-July 2022 was conducted in CINAHL, PubMed, SCOPUS, and Psych INFO. Search terms were developed in consultation with an experienced librarian and included combinations of key terms related to SMI and HRU measures. The search was conducted by the lead reviewer and then validated independently by the three co-authors.

Selection Criteria

Studies were included if they represented primary research of Medicaid-insured patients published in English January 2017-July 2022. They focused on adults age 18 and older, and were consistent with previous studies of Medicaid-insured patients with SMI (Pesa et al., 2015). Included studies were composed of patients with at least one SMI diagnosis (defined as major depressive disorder, bipolar disorder, schizo-

phrenia, and psychotic disorders) consistent with the National Institute of Mental Health (2024). Articles that addressed non-SMI mental illness such as anxiety in addition to SMI also were included. All articles evaluating measures of healthcare utilization were included (e.g., articles examining only a single metric of utilization such as 30-day readmissions). Systematic reviews and meta-analyses were excluded.

Quality Appraisal

Included studies were appraised critically by the lead reviewer using the National Institute of Health Quality Appraisal Tool for Observational Cohort and Cross-Sectional Studies (National Heart, Lung, and Blood Institute, 2021). Appraisal decisions were reviewed by co-authors for consistency. A study with a "Good" quality rating had a clearly stated objective and a uniform, pre-specified study population. All "Good" studies justified their sample size and effect estimates, used valid and reliable measures, and included confounding variable adjustment in their final analysis. A data extraction tool (see Table 1) was designed by the lead reviewer and extracted data were cross-checked by the co-reviewers.

Results

The search yielded 508 unique studies. Of these, 498 studies did not meet inclusion criteria, resulting in a sample of 10 studies (see Figure 1). All studies received a "Good" quality rating and were included in the final analysis. The included studies used large databases such as the Truven MarketScan Databases (Broder et al., 2018; Shrestha et al., 2020) and IBM MarketScan Commercial & Medicare Supplemental Databases (Zhdanava et al., 2022). One study used Transformed Medicaid Statistical Information System Files (Patel et al., 2022), another used MarketScan Medicaid Multi-State Database (Cook et al., 2020), and two studies used Kaiser Permanente Northern California, a large healthcare system databank (Banerjee et al., 2021; Geissler et al., 2021).

Insurance Payer Types and SMIs

Five studies included only Medicaid data (Banerjee et al., 2021; Becker et al., 2017; Cook et al., 2020; Patel et al., 2022; Pilon et al., 2019). Two included Medicaid and commercial payers (Geissler et al., 2021; Marcus et al., 2017), and the remaining three studies included all three payer types (Broder et al., 2018; Shrestha et al., 2020; Zhdanava et al, 2022). All four SMIs were represented in this review. Three articles included all four SMIs (Becker et al., 2017; Cook et al., 2020; Geissler et al., 2021). Two articles included schizophrenia and bipolar disorder only (Broder et al., 2018; Marcus et al., 2017), and one article focused only on major depression, bipolar disorder, and schizophrenia (Banerjee et al., 2021). Of the remaining four studies, one focused exclusively on schizophrenia (Patel et al., 2022) and three focused solely on types of major depression including treatment-resistant depression and major depression with suicidal ideation (Pilon et al., 2019; Shrestha et al., 2020; Zhdanava et al., 2022).

Distribution of HRU

The distribution of HRU metrics ranged from one study focused on healthcare costs (Banerjee et al., 2021) to five articles focused on hospitalizations, and emergency department (ED) and outpatient visits (Broder et al., 2018; Patel et al., 2022; Pilon et al., 2019; Shrestha et al., 2020; Zhdanava et al., 2022). Two studies focused on 30-day readmissions (Becker et al., 2017; Cook et al., 2020); three included rates of follow-up visits (Geissler et al., 2021; Marcus et al., 2017; Patel et al., 2022) and two included rates of pharmacotherapy and psychotherapy usage (Patel et al., 2022; Pilon et al., 2019).

Healthcare Costs

To identify medical diagnoses associated with persistent high HRU, Banerjee and co-authors (2021) isolated a high-utilizer cohort comprised of more than 1,400 Medicaid-insured adults who were the top 5% spenders in terms of inpatient and outpatient healthcare costs during their first year of Medicaid managed care enrollment. Medicaid spending for patients with SMI was four times higher than for those without it. Compared to patients with depression and bipolar disorder, only patients diagnosed with schizophrenia had a higher risk (1.5, 95% CI, 1.20-1.86) of having persistently high HRU going into the second year of Medicaid managed care enrollment.

Hospitalizations, ED Visits, and Outpatient Visits

Patel and co-authors (2022) conducted a nationwide cross-sectional analysis of HRU among 688,437 Medicaid-insured patients with schizophrenia (including 387,844 non-dual eligible patients). Findings indicated a national average of 34% of this group of patients had one or more inpatient admissions, 45% had one or more ED visits, and 86% had one or more outpatient visits during the study period. In a similar smaller multi-state study investigating HRU among 63,725 Medicaid-insured patients with schizophrenia and bipolar disorder, Broder and col-

TABLE 1.
Evidence Extraction Tool

Author, Year	Insurance Type	Sample	Study Design	Healthcare Resource Utilization (HRU) Outcomes	Serious Mental Illness (SMI)	Results
Pilon et al., 2019	Only Medicaid	26.6 million persons age ≥18	Retrospective longitudinal-matched cohort	Hospitalizations, ED visits, outpatient visits, pharmacotherapy psychotherapy usage	Major depression	Higher rates of all-cause HRU, higher PPPY healthcare costs
Cook et al., 2020	Only Medicaid	Ages 18-64, not dual Medicare-Medicaid	Retrospective cohort	30-day readmission	Major depression, schizophrenia, psychotic disorder, bipolar disorder	Higher readmission rates for patients with SMI
Patel et al., 2022	Only Medicaid	Patients with dual eligibility; claims not submitted to Medicaid not observed	Cross-sectional ecological	Hospitalizations, ED visits, outpatient visits, follow-up visits, pharmacotherapy/ psychotherapy usage	Schizophrenia	22% of patients had outpatient visit within 30 days of discharge.
Becker et al., 2017	Only Medicaid	Age ≥18 with at least one inpatient hospital episode for non-behavioral health condition before and following index hospitalization episode	Retrospective cohort	30-day readmission	Major depression, schizophrenia, psychotic disorder, bipolar disorder	Participants with SMI at higher risk of early rehospitalization
Zhdanava et al., 2022	All payers	Adult patients with one diagnosis for specific SMI	Retrospective observational matched cohort	Hospitalizations, ED visits, outpatient visits	Bipolar disorder, dementia, schizophrenia and schizoaffective disorder, substance-induced mood disorders	MDD cohort with suicidal ideation had 67.2 times more inpatient admissions, 8.9 times more ED visits, and 2.9 times more outpatient visits than the non-MDD cohort.
Shrestha et al., 2020	All payers	Age ≥18 with major depressive disorder diagnosis	Retrospective observational	Hospitalizations, ED visits, outpatient visits	Treatment-resistant depression	Patients with treatment-resistant depression had higher HRU, costs.

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TABLE 1. (CONTINUED)
Evidence Extraction Tool

Author, Year	Insurance Type	Sample	Study Design	Healthcare Resource Utilization (HRU) Outcomes	Serious Mental Illness (SMI)	Results
Geissler at al., 2021	All payers	Ages 21-64	Retrospective observational	Follow-up visits	Major depression, schizophrenia, psychotic disorder, bipolar disorder	Medicaid patients less likely to receive follow-up visits; visits were less effective in reducing ED visits/readmissions.
Marcus et al., 2017	Medicaid and commercial	Ages 18-64	Retrospective observational cohort	Follow-up visits	Schizophrenia, bipolar disorder	Medicaid patients less likely to receive follow-up visits; those who received follow-up visits had only slightly lower odds of readmission.
Broder et al., 2018	Medicaid and Medicare	Age ≥18 on index date with continuous health plan enrollment for 1 year	Retrospective observational cohort	Hospitalizations, ED visits, outpatient visits	Schizophrenia, bipolar disorder	Higher HRU and healthcare costs in patients with a diagnosis of both compared with patients with either diagnosis alone.
Banerjee et al., 2021	Only Medicaid	Ages 18-64 newly enrolled	Retrospective cohort	Healthcare costs	Major depression, bipolar disorder, schizophrenia	Persons with schizophrenia had higher risk of being persistent high utilizers.

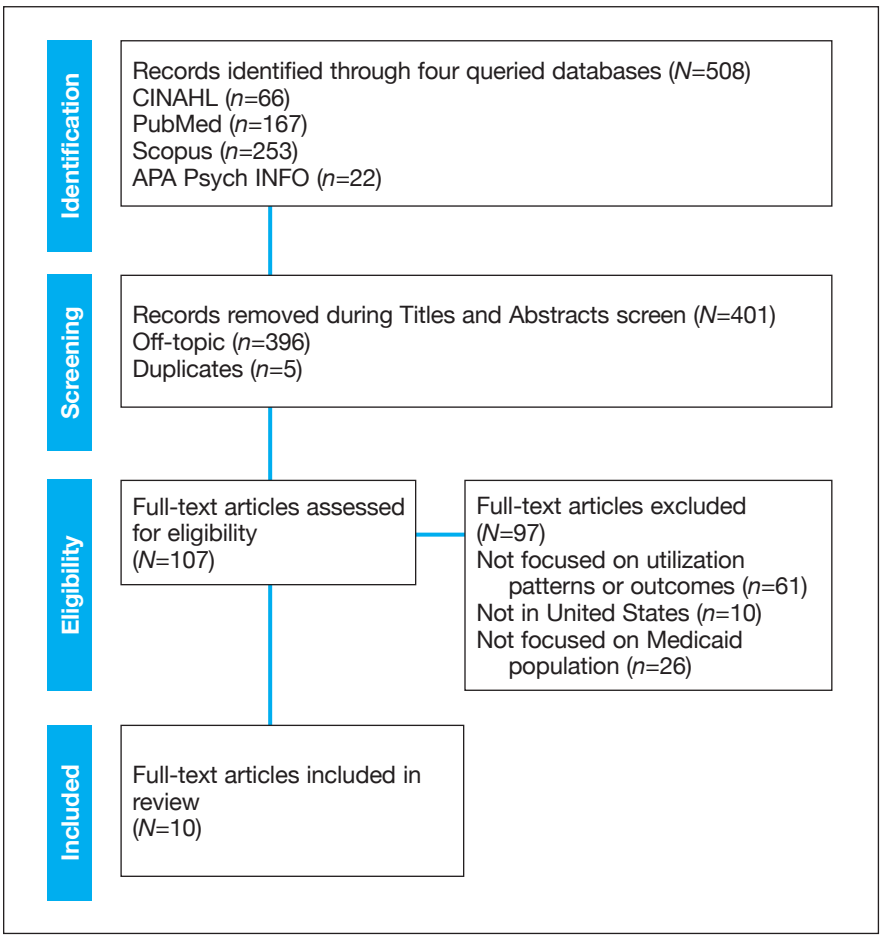
ED=emergency department, MDD=major depressive disorder, PPPY=per patient per year

leagues (2018) found 67.4% of Medicaid-insured individuals diagnosed with both disorders had an inpatient admission, along with 39.5% of patients with schizophrenia alone and 33.7% of patients with bipolar disorder alone. Patients with both conditions also had an average of 3.44 ED visits per year; those with schizophrenia alone had an average of 1.39 visits and those with bipolar disorder alone had an average of 1.29 visits. Zhdanava and co-authors (2022), Pilon and colleagues (2019), and Shrestha and co-authors (2020) reported elevated rates of inpatient, ED, and outpatient visits in Medicaid-insured patients with major depressive disorder, as well as populations with suicidal ideation and treatment-resistant depression.

30-Day Readmissions

In a study using claims data from 43,817 Medicaid-insured patients, Cook and co-authors (2020) found Medicaid-insured persons diagnosed with schizophrenia, bipolar disorder, or depression had a significantly greater risk of an unplanned 30-day readmission than those without SMI. Schizophrenia carried the greatest risk of 30-day readmission with an adjusted odds ratio of 1.46, followed by bipolar disorder at 1.25 and major depression at 1.18. Similarly, in a study using 1,956,169 hospitalization episodes spanning 8 years (2003-2011), Becker and colleagues (2017) found Medicaid patients diagnosed with bipolar disorder,

FIGURE 1.
PRISMA Diagram



psychotic disorder, and major depression had a 6%, 10%, and 17% higher risk, respectively, of being readmitted within 30 days of an index hospitalization compared to Medicaid patients without these disorders.

Post-Acute Follow-Up Visits

In their analysis over a calendar year, Patel and colleagues (2022) found only 22% of Medicaid-insured adults with schizophrenia had any outpatient visits (including primary care and behavioral health visits) within 30 days post-discharge, with percentages ranging from 58% in Washington, DC, to 8% in Colorado. Geissler and colleagues (2021) looked specifically at the relationships between receiving follow-up care and early readmission among Medicaid-insured patients with SMI and noted 33% received a post-acute follow-up visit. Marcus and co-authors (2017) conducted their analysis among Medicaid- and commercial-insured patients with schizophrenia and bipolar disorder to isolate risk factors associated with not receiving a mental health follow-up visit within 30 days after discharge. They found participation in a mental health follow-up visit was associated with being commercially insured, having longer inpatient stays, and more recent visits to a mental healthcare professional for both schizophrenia and bipolar patients.

Pharmacotherapy and Psychotherapy Usage

Medicaid-insured patients with SMI had low incidence of psychotherapy and pharmacotherapy utilization (Patel et al., 2022; Zhdanava et al., 2022). Similar to results in Patel and colleagues' (2022) study of 219,726 commercial- and Medicaid-insured patients living with major depression with suicidal ideation, Zhdanava and co-authors (2022) reported almost 50% of patients had no claims for antidepressants and 75% had no claims for psychotherapy visits. Concerns related to stigma, trust, and affordability were identified as potential barriers to accessing psychotherapy and pharmacotherapy services for Medicaid-insured patients with SMI.

Discussion and Implications for Practice, Education, and Research

This review identified disparate patterns in HRU among Medicaid patients with SMI. These patients experienced higher healthcare costs and HRU, including increased rates of inpatient hospitalizations, ED visits, outpatient visits, and 30-day readmissions. However, they had decreased rates of guideline-recommended care, such as 30-day follow-up visits, psychotherapy visits, and pharmacotherapy usage.

High healthcare costs are a potential indicator of a high disease burden and may be partly explained by the high prevalence of comorbid physical illnesses among patients with SMI (Wittbrodt et al., 2020). These patients have higher rates of chronic illnesses, such as hypertension, heart disease, obesity, diabetes, gastrointestinal disorders, and respiratory disorders (Pizzol et al., 2023). They also experience greater barriers to self-

management (Balogun-Katung et al., 2021). Symptoms associated with mental illness, such as avolition, alogia, delusions, psychosis, and psychomotor retardation, also affect patients' ability to engage in healthcare maintenance and prevention behaviors (Coventry et al., 2021).

Similarly, while an increase in outpatient visits may be associated with a high burden of illness, high rates of hospitalization and ED visits are explained most likely by a lack of continuity of care among patients with SMI and the disjointed manner in which their complex mental, physical, and social needs are addressed. In an observational cohort study of 19,324 patients and 215 practices, increased continuity of care was associated with 23%-27% lower risk of unplanned hospitalization and 8%-11% lower risk of ED presentation for patients with SMI (Ride et al., 2019). This study demonstrated the collective need for well-coordinated mental, physical, and social health services to improve physical health outcomes and reduce hospitalizations among Medicaid-insured patients with SMI. In addition to higher rates of hospitalizations, these patients also had high rates of 30-day readmissions and post-acute ED visits; both are important indicators of the quality of care received by Medicaid patients with SMI in acute care settings.

Lower rates of 30-day readmission rates historically have been linked to better nurse staffing and work environments (Kutney-Lee et al., 2015). A high rate of 30-day readmission also might signal unmet healthcare needs (Exuzides et al., 2022). Patients with SMI who received follow-up visits had increased healthcare costs and a higher chance of being rehospitalized, but a decreased probability of having a post-acute ED visit (Geissler et al., 2021). Patients who received follow-up visits were evaluated and referred to appropriate healthcare resources to meet their needs, including planned rehospitalization if necessary (Coleman et al., 2020). Post-hospital follow-up visits allow providers and patients to review reasons for hospitalization, discuss any unresolved or ongoing health issues, and identify any signs indicating a need for rehospitalization.

Two studies noted only marginal benefits for reducing hospitalization and ED visits for Medicaid patients with SMI who received a post-acute follow-up visit (Geissler et al., 2021; Marcus et al., 2017). While unexpected, this evidence suggests follow-up care received by Medicaid patients with SMI may be of lower quality or not as effective for these patients compared to SMI patients with commercial insurance (Geissler et al., 2021; Marcus et al., 2017). Medicaid patients with SMI also had lower rates of psychotherapy and pharmacotherapy usage (Patel et al., 2022; Pilon et al., 2019). These disparities signal greater socioeconomic barriers in accessing and receiving psychotherapy and pharmacotherapy for these patients. Growing evidence suggests a strong relationship between social determinants of health and medication usage among this population (Li et al., 2022). Socioeconomic disadvantage also is associated with decreased access to psychotherapy treatment (Niemeyer & Knaevelsrud, 2023). Medicaid patients face socioeconomic difficulties with scheduling and getting to their appointments, including challenges with timing, transportation, and childcare, as well as concerns about copayments (Jensen et al., 2020; Tambling et al., 2022).

None of the included articles highlighted the pivotal roles nurses play in improving health outcomes for Medicaid patients with SMI. However, findings clearly indicate a need for nursing interventions, such as enhanced discharge planning, transitional care planning, improved coordination of care, non-pharmaceutical behavioral support, and assistance with accessing pharmacotherapy and psychotherapy. Registered nurses in the medical-surgical setting are positioned uniquely to meet comprehensive care needs of Medicaid-insured patients with SMI, yet it is unclear if their training provides them with necessary skills to engage with and provide physical and mental health care to patients with mental illness (Avery et al., 2020).

Given the critical role medical-surgical nurses play in addressing unique needs of Medicaid-insured patients with SMI, they should receive thorough training and sufficient organizational resources. Nurse educators must ensure hospital nurses are well-prepared to interact with and treat patients with mental illness (Gu et al., 2021). Evidence shows nursing education programs incorporating mental health clinical placements increase student nurses' confidence in caring for individuals with mental illness (Stuhlmiller & Tolchard, 2019). Importantly, community mental health placements were shown to foster positive attitudinal changes toward patients with mental illness. In addition to training, medical-surgical nurses must be supported by robust organizational resources, including adequate staffing, an emphasis on baccalaureate nursing education, and a supportive work environment (Lake et al., 2022). High-quality organizational nursing resources are vital for nurses to deliver optimal patient care, including individualized care plans, effective discharge and transitional care planning, coordinated care with

interprofessional teams, behavioral support advocacy, and communication facilitation between patients and other healthcare providers (Kowalski et al., 2020; Lasater et al., 2021).

Nurse-led case management programs also have been shown to be effective in increasing rates of pharmacotherapy and psychotherapy usage in persons with SMI (Boyle & Schwinck, 2022; Liu et al., 2024). Nurse-led interventions, such as patient care management, medication management, and targeted patient education, were proven to be effective in increasing medication usage among patients with SMI (Balikai et al., 2022; Ghatwal et al., 2024; Liu et al., 2024). By providing targeted patient teaching around antipsychotic medications and increased post-discharge monitoring, nurses significantly increase the rates of pharmacotherapy use (Sieben et al., 2019). Nurse practitioners (NPs) also play an important role in improving access to quality physical and mental care for patients with SMI through implementing integrated care models and advanced case management interventions (Shepherd-Banigan et al., 2022). Effective coordination between psychiatric mental health NPs and primary care NPs in particular can increase access to and engagement in primary care for patients with SMI (Grove et al., 2021).

Limitations

Limitations inherent in design of systematic reviews include biases transferred from the original authors' work and potential publication bias based on the available literature surrounding the topic of healthcare utilization patterns of patients with SMI (Ayorinde et al., 2020). Data on important outcomes such as how many Medicaid-insured patients with SMI see a therapist regularly and how many receive prescriptions for antipsychotic medication, especially compared to commercially insured patients, was sparse in this current review. Limited data were available on racial or ethnic inequities in HRU patterns among Medicaid-insured patients with SMI. Some articles included other psychological illnesses that are not the focus of the current review, such as anxiety, dementia, substance use disorder, and alcohol use disorder.

Finally, authors did not address how having both an SMI and a substance use disorder can affect HRU patterns among Medicaid-insured patients. Future work should focus on the relationship between comorbid substance use disorder and HRU patterns among Medicaid-insured patients with SMI. Further investigation into health disparities among Medicaid-insured patients with SMI in acute and community settings is also necessary to ascertain how nurses might tailor interventions to address their unique healthcare needs.

Conclusion

Medicaid-insured patients with SMI face higher healthcare costs, increased hospitalizations, ED visits, outpatient visits, and 30-day readmissions while experiencing lower rates of psychotherapy and pharmacotherapy. Addressing these disparities requires coordinated, tailored care that considers social determinants of health and high-quality nursing interventions critical in reducing hospitalizations and ED visits. Improved nurse education, supportive practice environments, and enhanced coordination between psychiatric mental health and primary care NPs in community settings can help improve health outcomes for these patients. MSN

REFERENCES

Alcala, H.E., Ng, A.E., Gayen, S., & Ortega, A.N. (2020). Insurance types, usual sources of health care, and perceived discrimination. *Journal of the American Board of Family Medicine*, *33*(4), 580-591. https://doi.org/10.3122/jabfm.2020.04.190419

American Nurses Association. (2018). *The nurse's role in addressing discrimination: Protecting and promoting inclusive strategies in practice settings, policy, and advocacy* [Position statement]. https://www.nursingworld.org/globalassets/practiceandpolicy/nursing-excellence/ana-position-statements/social-causes-and-health-care/the-nurses-role-in-addressing-discrimination.pdf

Avery, J., Schreier, A., & Swanson, M. (2020). A complex population: Nurse's professional preparedness to care for medical-surgical patients with mental illness. *Applied Nursing Research*, 52, 151232. https://doi.org/10.1016/j.apnr.2020.151232

Ayorinde, A.A., Williams, I., Mannion, R., Song, F., Skrybant, M., Lilford, R.J., & Chen, Y.-F. (2020). Assessment of publication bias and outcome reporting bias in systematic reviews of health services and delivery research: A meta-epidemiological study. *PLoS One, 15*(1), e0227580. https://doi.org/10.1371/journal.pone.0227580

Balikai, S.I., Rentala, S., Mudakavi, I.B., & Nayak, R.B. (2022). Impact of nurse-led medication adherence therapy on bipolar affective disorder: A randomized controlled trial. *Perspectives in Psychiatric Care*, *58*(4), 2676-2686. https://doi.org/10.1111/ppc.13108

Balogun-Katung, A., Carswell, C., Brown, J.V.E., Coventry, P., Ajjan, R., Alderson, S., ... Taylor, J. (2021). Exploring the facilitators, barriers, and strategies for self-management in adults living with severe mental illness, with and without long-term conditions: A qualitative evidence synthesis. *PLoS ONE*, *16*(10), e0258937. https://doi.org/10.1371/journal.pone.0258937

Banerjee, S., Alabaster, A., Kipnis, P., & Adams, A.S. (2021). Factors associated with persistent high health care utilization in managed Medicaid. *The American Journal of Managed Care*, 27(8), 340-344. https://doi.org/10.37765/ajmc.2021.88725

Becker, M.A., Boaz, T.L., Andel, R., & Hafner, S. (2017). Risk of early rehospitalization for non-behavioral health conditions among adult Medicaid beneficiaries with severe mental illness or substance use disorders. *The Journal of Behavioral Health Services & Research*, *44*, 113-121. https://doi.org/10.1007/s11414-016-9516-9

Boyle, M.K., & Schwinck, J. (2022). Reducing missed psychotherapy appointments: An advanced practice nurse-initiated telephone orientation protocol. *Perspectives in Psychiatric Care*, *58*(4), 2756-2763. https://doi.org/10.1111/ppc.13116

Broder, M.S., Greene, M., Chang, E., Hartry, A., Touya, M., Munday, J., & Yan, T. (2018). Health care resource use, costs, and diagnosis patterns in patients with schizophrenia and bipolar disorder: Real-world evidence from US claims databases. *Clinical Therapeutics*, *40*(10), 1670-1682. https://doi.org/10.1016/j.clinthera.2018.08.004

Brom, H., Anusiewicz, C.V., Udoeyo, I., Chittams, J., & Brooks Carthon, J.M. (2022). Access to post-acute care services reduces emergency department utilisation among individuals insured by Medicaid: An observational study. *Journal of Clinical Nursing*, *31*(5-6), 726-732. https://doi.org/10.1111/jocn.15932

Brooks Carthon, M., Brom, H., McHugh, M., Sloane, D.M., Berg, R., Merchant, R., ... Aiken, L.H. (2021). Better nurse staffing is associated with survival for black patients and diminishes racial disparities in survival after in-hospital cardiac arrests. *Medical Care*, *59*(2), 169-176. https://doi.org/10.1097/MLR.0000000000001464

Coleman, K.J., Dreskin, M., Hackett, D.L., Aunskul, A., Liu, J., Imley, T.M., ... Beaubrun, G.F. (2020). A roadmap for institutionalizing collaborative care for depression in a large integrated healthcare system. *Journal of General Internal Medicine*, *35*(Suppl 2), 839-848. https://doi.org/10.1007/s11606-020-06102-8

Cook, J.A., Burke-Miller, J.K., Jonikas, J.A., Aranda, F., & Santos, A. (2020). Factors associated with 30-day readmissions following medical hospitalizations among Medicaid beneficiaries with schizophrenia, bipolar disorder, and major depressive disorder. *Psychiatry Research*, *291*, 113168. https://doi.org/10.1016/j.psychres.2020.113168

Coventry, P.A., Young, B., Balogun-Katang, A., Taylor, J., Brown, J.V.E., Kitchen, C., ... Siddiqi, N. (2021). Determinants of physical health self-management behaviours in adults with serious mental illness: A systematic review. *Frontiers in Psychiatry*, *12*, 723962. https://doi.org/10.3389/fpsyt.2021.723962

Exuzides, A., To, T.M., Abbass, I.M., Ta, J.T., Patel, A.M., Surinach, A., ... Luo, J. (2022). Healthcare resource utilization and costs in individuals with Huntington's disease by disease stage in a US population. *Journal of Medical Economics*, *25*(1), 722-729. https://doi.org/10.1080/13696998.2022.2076997

Gandhi, A.B., Onukwugha, E., Albarmawi, H., Johnson, A., Myers, D.E., Gray, D., ... Shulman, L.M. (2021). Health care resource utilization associated with Parkinson disease among Medicare beneficiaries. *Neurology*, *97*(6), e597-e607. https://doi.org/10.1212/WNL.0000000000012290

Geissler, K.H., Cooper, M.I., & Zeber, J.E. (2021). Association of follow-up after an emergency department visit for mental illness with utilization based outcomes. *Administration and Policy in Mental Health*, *48*(4), 718-728. https://doi.org/10.1007/s10488-020-01106-2

Ghatwal, H., Joseph, J., & Jangid, P. (2021). Effect of nurse-led screening linked brief psycho-education for improving adherence to antipsychotic medications among clients with mental illness: A quasi-experimental study. *Journal of Mental Health and Human Behavior*, *26*(1), 28-35. https://doi.org/10.4103/jmhhb.jmhhb_175_20

Gross, C.P., Meyer, C.S., Ogale, S., Kent, M., & Wong, W.B. (2022). Associations between Medicaid insurance, biomarker testing, and outcomes in patients with advanced NSCLC. *Journal of the National Comprehensive Cancer Network*, *20*(5), 479-487. https://doi.org/10.6004/jnccn.2021.7083

Grove, L.R., Gertner, A.K., Swietek, K.E., Lin, C.-C., Ray, N., Malone, T.L., ... Steiner, B.D. (2021). Effect of enhanced primary care for people with serious mental illness on service use and screening. *Journal of General Internal Medicine*, *36*(4), 970-977. https://doi.org/10.1007/s11606-020-06429-2

Gu, L., Jiao, W., Xia, H., & Yu, M. (2021). Psychiatric-mental health education with integrated role-play and real-world contact can reduce the stigma of nursing students towards people with mental illness. *Nurse Education in Practice*, *52*, 103009. https://doi.org/10.1016/j.nepr.2021.103009

Hall, J. P., LaPierre, T.A., & Kurth, N.K. (2019). Medicaid managed care: Issues for enrollees with serious mental illness. *The American Journal of Managed Care*, *25*(9), 450-456. https://doi.org/10.37765/ajmc.2019.43016

Hill, L., Ndugga, N., & Artiga, S. (2025, February 13). *Health coverage by race and ethnicity, 2010-2023*. KFF. https://www.kff.org/racial-equity-and-health-policy/issue-brief/health-coverage-by-race-and-ethnicity

Hwong, A.R., Chagwedera, D.N., Thomas, M., Niu, G., Quan, J., Vittinghoff, E., ... Mangurian, C. (2022). CRANIUM: A quasi-experimental study to improve metabolic screening and HIV testing in community mental health clinics compared to usual care. *BMC Psychiatry*, *22*, Article 687. https://doi.org/10.1186/s12888-022-04293-4

Jensen, E.J., Wieling, E., & Mendenhall, T. (2020). A phenomenological study of clinicians' perspectives on barriers to rural mental health care. *Journal of Rural Mental Health*, *44*(1), 51-61. https://doi.org/10.1037/rmh0000125

Kaiser Family Foundation. (2023). *Medicaid coverage rates for people ages 0-64 by race/ethnicity.* https://www.kff.org/medicaid/state-indicator/people-0-64-medicaid-rate-by-raceethnicity

Kowalski, M.O., Basile, C., Bersick, E., Cole, D.A., McClure, D.E., & Weaver, S.H. (2020). What do nurses need to practice effectively in the hospital environment? An integrative review with implications for nurse leaders. *Worldviews on Evidence-Based Nursing, 17*(1), 60-70. https://doi.org/10.1111/wvn.12401

Kutney-Lee, A., Stimpfel, A.W., Sloane, D.M., Cimiotti, J.P., Quinn, L.W., & Aiken, L.H. (2015). Changes in patient and nurse outcomes associated with Magnet hospital recognition. *Medical Care*, *53*(6), 550-557. https://doi.org/10.1097/MLR.0000000000000355

Lake, E.T., Riman, K.A., & Lee, C.S. (2022). The association between hospital nursing resource profiles and nurse and patient outcomes. *Journal of Nursing Management*,

30(3), 836-845. https://doi.org/10.1111/jonm.13553

Lasater, K.B., McHugh, M., Rosenbaum, P.R., Aiken, L.H., Smith, H., Reiter, J.G., ... Silber, J.H. (2021). Valuing hospital investments in nursing: Multistate matched-cohort study of surgical patients. *BMJ Quality & Safety*, *30*(1), 46-55. https://doi.org/10.1136/bmjqs-2019-010534

Li, N.C., Alcusky, M., Masters, G.A., & Ash, A.S. (2022). Association of social determinants of health with adherence to second-generation antipsychotics for people with bipolar disorders in a Medicaid population. *Medical Care*, *60*(2), 106-112. https;//doi.org/10.1097/MLR.0000000000001670

Liu, W.-I, Hsieh, W.-L., Lai, C.-T., Liu, C.-C., Tai, Y.-M., & Liu, C.-Y. (2024). Effectiveness of a needs-tailored nurse-led recovery program for community-dwelling people with schizophrenia: A cluster-randomized controlled trial. *BMC Nursing*, *23,* Article 329. https://doi.org/10.1186/s12912-024-01986-x

Marcus, S.C., Chuang, C.-C., Ng-Mak, D.S., & Olfson, M. (2017). Outpatient follow-up care and risk of hospital readmission in schizophrenia and bipolar disorder. *Psychiatric Services*, *68*(12), 1239-1246. https://doi.org/10.1176/appi.ps.201600498

McConnell, K.J., Charlesworth, C.J., Zhu, J.M., Meath, T.H.A., George, R.M., Davis, M.M., ... Kim, H. (2020). Access to primary, mental health, and specialty care: A comparison of Medicaid and commercially insured populations in Oregon. *Journal of General Internal Medicine*, *35*, 247-254. https://doi.org/10.1007/s11606-019-05439-z

McMaughan, D.J., Oloruntoba, O., & Smith M.L. (2020). Socioeconomic status and access to healthcare: Interrelated drivers for healthy aging. *Frontiers in Public Health, 8*, 231. https://doi.org/10.3389/fpubh.2020.00231

Medicaid and CHIP Payment and Access Commission. (2021). Access to mental health services for adults covered by Medicaid. In *Report to Congress on Medicaid and CHIP* (pp. 31-77). https://www.macpac.gov/wp-content/uploads/2021/06/June-2021-Report-to-Congress-on-Medicaid-and-CHIP.pdf

Mongelli, F., Georgakopoulos, P., & Pato, M.T. (2020). Challenges and opportunities to meet the mental health needs of underserved and disenfranchised populations in the United States. *Focus*, *18*(1), 16-24. https://doi.org/10.1176/appi.focus.20190028

National Academies of Sciences, Engineering, and Medicine. (2021). *The future of nursing 2020-2030: Charting a path to achieve health equity*. The National Academies Press. https://doi.org/10.17226/25982

National Heart, Lung, and Blood Institute. (2021, July). *Study quality assessment tools: Quality assessment tool for observational cohort and cross-sectional studies*. https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools

National Institute of Mental Health. (2024). *Mental illness*. https://www.nimh.nih.gov/health/statistics/mental-illness

Niemeyer, H., & Knaevelsrud, C. (2023). Socioeconomic status and access to psychotherapy. *Journal of Clinical Psychology*, *79*(4), 937-953. https://doi.org/10.1002/jclp.23449

Nong, P., Raj, M., Creary, M., Kardia, S.L.R., & Platt, J.E. (2020). Patient-reported experiences of discrimination in the US health care system. *JAMA Network Open*, *3*(12), e2029650. https://doi.org/10.1001/jamanetworkopen.2020.29650

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *International Journal of Surgery*, *88*, 105906. https://doi.org/10.1016/j.ijsu.2021.105906

Patel, C., Pilon, D., Gupta, D., Morrison, L., Lafeuille, M.-H., Lefebvre, P., & Benson, C. (2022). National and regional description of healthcare measures among adult Medicaid beneficiaries with schizophrenia within the United States. *Journal of Medical Economics*, *25*(1), 792-807. https://doi.org/10.1080/13696998.2022.2084234

Pesa, J.A., Muser, E., Montejano, L.B., Smith, D.M., & Meyers, O.I. (2015). Costs and resource utilization among Medicaid patients with schizophrenia treated with paliperidone palmitate or oral atypical antipsychotics. *Drugs - Real World Outcomes*, *2*(4), 377-385. https://doi.org/10.1007/s40801-015-0043-4

Pilon, D., Sheehan, J.J., Szukis, H., Singer, D., Jacques, P., Lejeune, D., ... Greenberg, P.E. (2019). Medicaid spending burden among beneficiaries with treatment-resistant depression. *Journal of Comparative Effectiveness Research*, *8*(6), 381-392. https://doi.org/10.2217/cer-2018-0140

Pizzol, D., Trott, M., Butler, L., Barnett, Y., Ford, T., Neufeld, S.A., ... Smith, L. (2023). Relationship between severe mental illness and physical multimorbidity: A meta-analysis and call for action. *BMJ Mental Health*, *26*(1), e300870. https://doi.org/10.1136/bmjment-2023-300870

Rajeswaren, V., Lu, V., Chen, H., Patnaik, J.L., & Manoharan, N. (2024). Healthcare resource utilization and costs in an at-risk population with diabetic retinopathy. *Translational Vision Science & Technology*, *13*(2), 12. https://doi.org/10.1167/tvst.13.2.12

Recchia, D.R., Cramer, H., Wardle, J., Lee, D.J., Ostermann, T., & Lauche, R. (2022). Profiles and predictors of healthcare utilization: Using a cluster-analytic approach to identify typical users across conventional, allied and complementary medicine, and self-care. *BMC Health Services Research*, *22*, Article 29. https://doi.org/10.1186/s12913-021-07426-9

Ride, J., Kasteridis, P., Gutacker, N., Doran, T., Rice, N., Gravelle, H., ... Jacobs, R. (2019). Impact of family practice continuity of care on unplanned hospital use for people with serious mental illness. *Health Services Research*, *54*(6), 1316-1325. https://doi.org/10.1111/1475-6773.13211

Ronaldson, A., Elton, L., Jayakumar, S., Jieman, A., Halvorsrud, K., & Bhui, K. (2020). Severe mental illness and health service utilisation for nonpsychiatric medical disorders: A systematic review and meta-analysis. *PLoS Medicine*, *17*(9), e1003284. https://doi.org/10.1371/journal.pmed.1003284

Shepherd-Banigan, M., Drake, C., Dietch, J.R., Shapiro, A., Tabriz, A.A., Van Voorhees, E.E., ... Goldstein, K.M. (2022). Primary care engagement among individuals with experiences of homelessness and serious mental illness: An evidence map. *Journal of General Internal Medicine*, *37*(6), 1513-1523. https://doi.org/10.1007/s11606-021-07244-z

Shrestha, A., Roach, M., Joshi, K., Sheehan, J.J., Goutam, P., Everson, K., ... Jena, A.B. (2020). Incremental health care burden of treatment-resistant depression among com-

mercial, Medicaid, and Medicare payers. *Psychiatric Services*, *71*(6), 593-601. https://doi.org/10.1176/appi.ps.201900398

Sieben, A., van Onzenoort, H.A., van Dulmen, S., van Laarhoven, C., & Bredie, S.J. (2019). A nurse-based intervention for improving medication adherence in cardiovascular patients: an evaluation of a randomized controlled trial. *Patient Preference and Adherence, 13*, 837085 2. https://doi.org/10.2147/ppa.s197481

Singletary, K.A., & Chin, M.H. (2023). What should antiracist payment reform look like? *AMA Journal of Ethics*, *25*(1), E55-E65. https://doi.org/10.1001/amajethics.2023.55

Snowden, L., & Graaf, G. (2019). The "undeserving poor," racial bias, and Medicaid coverage of African Americans. *Journal of Black Psychology*, *45(*3), 130-142. https://doi.org/10.1177/0095798419844129

Stuhlmiller, C., & Tolchard, B. (2019). Understanding the impact of mental health placements on student nurses' attitudes towards mental illness. *Nurse Education in Practice*, *34*, 25-30. https://doi.org/10.1016/j.nepr.2018.06.004

Tambling, R., Hynes, K.C., & D'Aniello, C. (2022). Are barriers to psychotherapy treatment seeking indicators of social determinants of health? A critical review of the literature. *The American Journal of Family Therapy*, *50*(5), 443-458. https://doi.org/10.1080/01926187.2020.1867931

van Dusseldorp, L., Groot, M., van Vught, A., Goossens, P., Hulshof, H., & Peters, J. (2023). How patients with severe mental illness experience care provided by psychiatric mental health nurse practitioners. *Journal of the American Association of Nurse Practitioners*, *35*(5), 281-290. https://doi.org/10.1097/JXX.0000000000000867

Wittbrodt, E., Bhalla, N., Sundell, K.A., Gao, Q., Dong, L., Cavender, M.A., ... Mellstrom, C. (2020). Assessment of the high risk and unmet need in patients with CAD and type 2 diabetes (ATHENA): US healthcare resource utilization, cost and burden of illness in the Diabetes Collaborative Registry. *Endocrinology, Diabetes & Metabolism*, *3*(3), e00133. https://doi.org/10.1002/edm2.133

Yalcin, S.U., Bilgin, H., & Ozaslan, Z. (2019). Physical healthcare of people with serious mental illness: A cross-sectional study of nurses' involvement, views, and current practices. *Issues in Mental Health Nursing*, *40*(10), 908-916. https://doi.org/10.1080/01612840.2019.1619201

Zhdanava, M., Voelker, J., Pilon, D., Joshi, K., Morrison, L., Sheehan, J.J., ... Citrome, L. (2022). Excess healthcare resource utilization and healthcare costs among privately and publicly insured patients with major depressive disorder and acute suicidal ideation or behavior in the United States. *Journal of Affective Disorders*, *311*, 303-310. https://doi.org/10.1016/j.jad.2022.05.086

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